

Assignment:- 4

Unit:- 3

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Subject: Deep learning.

① What is an autoencoder in deep learning?

→ An autoencoder is a neural network used for unsupervised learning that learns to encode input data into a lower-dimensional representation and then decode it to reconstruct the original input. It is commonly used for dimensionality reduction, feature learning, and noise removal.

② What are the main components of an autoencoder?

→ The main components are:

- Encoder: Converts the input data into a compressed representation.
- Latent space (bottleneck): The compact representation that captures the most important features.
- Decoder: Reconstructs the original input from the latent representation.

③ How is an autoencoder related to PCA?

→ A linear autoencoder with a single hidden layer and mean squared error loss can learn the same subspace as principal component analysis (PCA). Both methods perform dimensionality reduction, but autoencoders can also model nonlinear relationships, while PCA is limited to linear transformations.

④ What is regularization in autoencoders?

→ Regularization in autoencoders refers to the technique that restricts the model from simply copying the input, forcing it to learn useful features. This is done by adding penalties or constraints such as sparsity, noise or weight penalties.

⑤ Define a denoising autoencoder.

→ A denoising autoencoder is a type of autoencoder that is trained to reconstruct the original clean input from a corrupted or noisy version of the input, helping the model learn robust feature representations.

⑥ What is a sparse autoencoder?

→ A sparse autoencoder is an autoencoder that uses a sparsity constraint on hidden units, meaning only a small number of neurons are active at a time. This helps the model learn more meaningful and distinct features.

⑦ What is meant by bias-variance tradeoff?

→ The bias-variance tradeoff describes the balance between bias (error due to overly simple models) and variance (error due to overly complex models that overfit the data). A good model finds the optimal balance to generalize well on new data.

⑧ What is L_2 regularization?

→ L_2 regularization is a technique that adds a penalty proportional to the square of the weights to the loss function. It discourages large weights and helps reduce overfitting, also known as weight decay.

(9) How does early stopping help prevent overfitting?

→ Early stopping monitors the model's performance on a validation dataset during training and stops training when validation error starts increasing, preventing the model from memorizing the training data.

(10) What is dataset augmentation?

→ Dataset augmentation is a technique used to increase the size and diversity of training data by creating modified versions of existing data, which helps improve model generalization.

(11) Give two examples of data augmentation techniques used in image data.

→ Two examples are:

- Image rotation

- Horizontal or vertical flipping.

⑫ What is an encoder-decoder model?

→ An encoder-decoder model is a neural network architecture where the encoder converts input data into a compact representation, and the decoder generates an output from that representation. It is commonly used in tasks like machine translation, text summarization, and image captioning.

⑬ What is the attention mechanism?

→ The attention mechanism is a technique that allows a model to focus on the most relevant parts of the input when generating each part of the output, rather than treating all input information equally.

(14) Why is attention important in deep learning models?

→ Attention improves performance by allowing models to capture long-range dependencies, focus on important information, and handles large sequences more effectively, especially in tasks like natural language processing and translation.

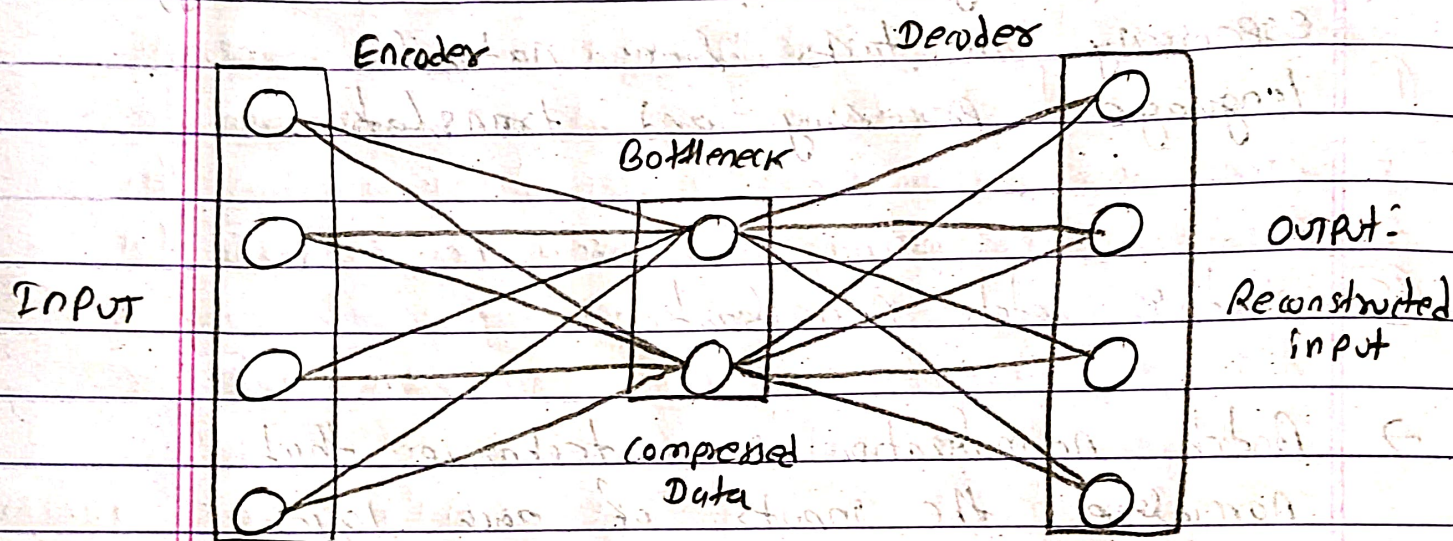
(15) What is batch normalization?

→ Batch normalization is a technique that normalizes the inputs of each layer during training by adjusting and scaling the activations. It helps speed up training, stabilize learning, and improve model performance.

Long Questions:

1) Explain the architecture and working of autoencoders with a neat diagram.

→ An autoencoder is a neural network designed to learn efficient representation of data by compressing input data and reconstructing it.



Architecture components

1) Input Layer

- Receives the original data x .

2) Encoder

- Transforms the input into a lower-dimensional representation

Function: $h = f(Wx + b)$

- h is the latent representation.

3) Latent space (Bottleneck layer)

- The compressed feature representation of the input.
- Forces the model to capture the most important patterns.

4) Decoder

- Reconstructs the input from the latent representation
- Function:

$$\hat{x} = g(w'h + b')$$

5) Output layer

produces the reconstructed input $\hat{x}$.

Working Process:

- 1) Input data is passed in the encoder
- 2) The encoder compresses it into latent features
- 3) The decoder reconstructs it into the original data
- 4) The model minimizes reconstruction error using a loss function like Mean Squared Error (MSE).

$$\text{Loss function} : L = \frac{1}{n} \sum (x - \hat{x})^2$$

② Discuss the relationship between autoencoders and PCA. How are they similar and different?

→ Principal Component Analysis (PCA) and autoencoders are both used for dimensionality reduction.

Similarities:

- i) Both reduce high-dimensional data into lower dimensions.
- ii) Both aim to retain maximum important information.
- iii) Both learn compressed representations of input data.

Differences

Feature	PCA	Autoencoder
Type	Statistical method	Neural method
mapping	Linear	Linear or Non-linear
layers	No hidden layers	uses multiple layers
training	Eigen decomposition	Backpropagation
flexibility	Limited	Highly flexible

key insight:

- A single-layer linear autoencoder with MSE loss learns the same subspace as PCA.

③ Explain regularization techniques used in autoencoders and their importance.

→ Regularization techniques prevent an autoencoder from simply memorizing the input.

Common Regularization techniques

1) L1/L2 weight regularization

- Adds penalties to large weights

$$\text{Loss} = \text{Reconstruction Error} + \lambda \|W\|^2$$

2) Sparse Regularization

↳ Forces most neurons to stay inactive.

↳ Encourages learning distinct features

3) Denoising Regularization

↳ Add noise to inputs and trains the network to recover original data.

4) Contractive Regularization

- Penalizes large gradients in encoder mapping

- Make representations robust to small changes.

Importance

Regularization helps to

- Prevent overfitting
- Improve generalization
- Learn meaningful feature representations
- Increase model robustness

④ Describe denoising autoencoders & explain how they improve feature learning.

→ A denoising autoencoders (DAE) is trained to reconstruct clean input from corrupted input.

Working process:

- I) Add noise to input data.
- II) Feed noisy data into encoder.
- III) Encoder extract latent features.
- IV) Decoder reconstructs original clean data.
- V) Training minimizes reconstruction error between clean and reconstructed output.

Example noise type:

- Gaussian Noise
- Random masking
- Salt-and-pepper noise

Benefits

- learns robust features
- Reduces sensitivity to noise
- Improves generalization
- Helps in image denoising and feature extraction

⑤ Explain sparse autoencoders and discuss the role of sparsity constraints.

→ A sparse autoencoder restricts the number of active neurons in hidden layers. Even if the hidden layer is large, only few neurons activate for each input.

Sparsity constraint

- usually enforced using KL divergence Penalty:

$$KL(P||\hat{P}) = P \log \frac{P}{\hat{P}} + (1-P) \log \frac{1-P}{1-\hat{P}}$$

where:

- P = desired sparsity level
- $\hat{P}$ = actual ~~sparsity level~~ neuron activation

~~total~~ loss function:

$$\text{loss} = \text{Reconstruction loss} + \lambda KL(P||\hat{P})$$

Advantages:

- Learns more interpretable features.
- Encourages features specialization.
- Improves representation learning.

Q Explain the bias - variance trade off and its significance in deep learning models.

→ The bias - variance trade off explains the balance between underfitting and overfitting.

Bias

- Bias is the error caused by overly simple models.
- Example: Linear model trying to fit nonlinear data
High bias $\rightarrow$ underfitting.

Variance

- Variance is the error caused by high model complexity.
- Ex: model memorizes training data.
High variance $\rightarrow$ overfitting.

Trade off

Case	Bias	Variance
- underfitting	High	Low
- overfitting	Low	High
- Good model	Balanced	Balanced

Goal

- Find the optimal balance that minimizes total error.

⑦ Explain L_2 regularization mathematically and discuss how it reduces overfitting.

→ L_2 regularization adds a penalty term proportional to the square of weights.

Regularization loss function: $J(w) = \text{loss} + \lambda \sum w^2$

where: • λ = regularization parameter

• w = model weights

Effect on gradient

During gradient descent: $w = w - \eta \left(\frac{\partial \text{loss}}{\partial w} + 2\lambda w \right)$

This shrinks weights toward zero.

Benefits

- Prevents very large weights
- Improves model generalization
- Reduces overfitting

Also called weight decay.

⑧ What is early stopping? Explain how it works during model training.

→ Early stopping is a regularization technique used to stop training before overfitting occurs.

Working:

- During training:
- i) Split dataset into training and validation sets.
 - ii) Train the model normally.
 - iii) Monitor validation loss.

Scenario:

Epoch	Training loss	Validation loss
1	High	High
10	Lower	Lower
20	Lower	Starts increasing

When validation loss starts increasing, training is stopped.

Benefits

- Prevents overfitting
- Saves training time
- Improves generalization.

⑨ Explain dataset augmentation and discuss its advantages in training deep learning models.

→ Dataset augmentation artificially increases the size and diversity of training data by applying transformations common in image, audio, and text data.

Image Augmentation Techniques

- Rotation
- Flipping
- Cropping
- Scaling
- Brightness adjustment
- Noise injection

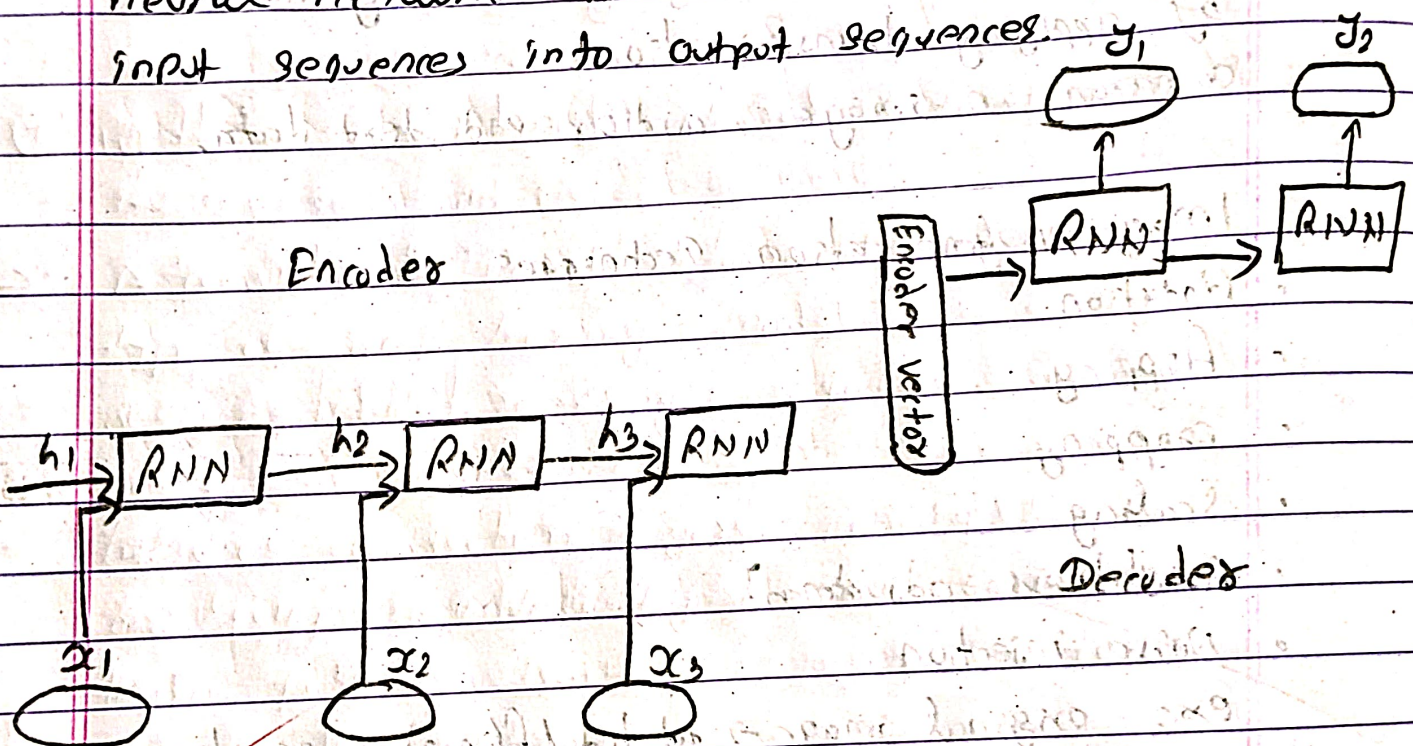
ex: original image → rotated / flipped → new training samples.

Advantages:

- i) increases dataset size
- ii) Reduces overfitting
- iii) Improves model generalization
- iv) makes model robust to variations
- v) Reduces need for large labeled datasets.

10) Explain the encoder-decoder architecture and mention its applications.

→ The encoder-decoder architecture is a neural network framework used to transform input sequences into output sequences.



Components

1) Encoder

- processes input sequence
 - convert it into context vector (hidden representation)
- Ex: Input sentence $\rightarrow$ encoder $\rightarrow$ vector representation

2) Context vector

- Encodes all information from input

3) Decoder

→ uses context vector to generate output sequence

ex: vector → decoder → translated sentence

* Working Example

Machine translation:

English: "How are you?"

Encoder → Context vector → Decoder → Spanish: "¿Cómo estás?"

Applications

- Machine Translation
- Text Summarization
- Image Captioning
- Speech Recognition
- Chatbots
- Question Answering

